

### HIGHER TIER ONLY QUESTIONS

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks available |          |          |          |          |          |
|----------|-----|------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 2        | (a) |      |  | $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$<br>award (1) for both reactants<br>award (1) for product <b>and</b> balancing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2               |          |          | 2        |          |          |
|          | (b) | (i)  |  | $\boxed{2} \text{Fe}_2\text{O}_3 + 3\text{C} \longrightarrow \boxed{4} \text{Fe} + \boxed{3} \text{CO}_2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 | 1        |          | 1        |          |          |
|          |     | (ii) |  | <u>iron oxide</u> is reduced $\Rightarrow$ it loses oxygen (1)<br>neutral answer – <u>iron</u> is reduced $\Rightarrow$ it loses oxygen<br>carbon is oxidised $\Rightarrow$ it gains oxygen (1)<br>if no other credit award (1) for generic description or for identifying which reactants are oxidised and reduced <ul style="list-style-type: none"> <li>• oxidation is gaining oxygen, reduction is losing oxygen</li> <li>• iron oxide is reduced (to form iron) and carbon is oxidised (to form carbon dioxide)</li> <li>• oxygen goes from iron oxide to carbon</li> </ul> reference to electrons is neutral |                 | 2        |          | 2        |          |          |
|          | (c) | (i)  |  | decomposition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1               |          |          | 1        |          | 1        |
|          |     | (ii) |  | $\text{CaSiO}_3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1               |          |          | 1        |          |          |
|          |     |      |  | <b>Question 2 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>4</b>        | <b>3</b> | <b>0</b> | <b>7</b> | <b>0</b> | <b>1</b> |